

Research Paper

Enhancing building energy efficiency across diverse climatic zones in marocco through optimized eco-friendly insulation thickness with air gap

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ARTICLE INFO

Keywords:

Life cycle cost analysis

CFD simulation

Thermal insulation

Optimal insulation thickness

Cooling and heating loads

Energy efficiency

ABSTRACT

The study addresses the growing imperative to reduce the environmental impact of construction by revisiting earth-based building materials, which offer both sustainability and local resource advantages. Focused on Morocco's unique climate, the research evaluates an innovative thermal insulation method for south-facing walls using eco-friendly materials. Its primary aim is to determine the optimal insulation thickness, crucial for regulating heating and cooling demands in diverse Moroccan climates, and to enhance it through a smart solution. Employing Life Cycle Cost Analysis (LCCA) and Computational Fluid Dynamics (CFD) modeling, the study compares the thermal performance of two wall configurations: ecological insulation alone versus with the integrated smart solution. Findings reveal the optimum insulation thickness, resulting in significant energy savings during both summer and winter. Moreover, the proposed intelligent solution further reduces insulation thickness, leading to energy savings of 3.72 % and 3.68 % respectively in summer and winter, alongside reduced insulation costs. The optimization strategy, maintaining internal temperatures at comfortable levels and modulating air velocity, aims to make substantial contributions towards reducing energy consumption and environmental impact in the construction sector.

1. Introduction

The building industry's reliance on energy has surged due to the rising demand for architectural comfort, leading to a significant increase in energy consumption. Heating and cooling currently constitute nearly 40 % of the energy used in this sector. Projections indicate that energy consumption in the construction sector will surge by almost 28 % by 2040. Furthermore, the widespread use of petrochemical insulation materials poses a threat to environmental sustainability by significantly elevating CO₂ emissions. Notably, in Morocco, the construction sector contributes to 33 % of the country's total energy consumption (IEA, [Buildings-Energy System, 2021](#)).

To address this pressing issue, Morocco has recently introduced new thermal regulations for buildings, known as RTCM. These regulations aim to reduce energy consumption and enhance energy efficiency, presenting a promising opportunity to lower energy expenditures in the construction industry. This move marks a significant stride towards

sustainability and a notable reduction in the sector's environmental footprint.

In the Mediterranean region, where the impact of climate and weather conditions on energy consumption is profoundly felt, architectural design plays a pivotal role in determining a building's energy and thermal performance. This unique environmental context necessitates the integration of passive techniques and advanced insulating materials into construction practices to achieve substantial energy savings.

Passive techniques, such as strategic building orientation, natural ventilation, and the use of thermal mass, have been extensively explored to optimize energy efficiency. These methods leverage the natural climate to reduce reliance on artificial heating and cooling systems, thereby decreasing overall energy consumption.

Simultaneously, researchers are tirelessly working on innovative solutions to enhance thermal insulation in buildings. Their efforts focus on developing materials and methods that not only improve insulation but also promote sustainability. These advancements aim to reduce the

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<https://doi.org/10.1016/j.egy.2024.07.040>

Received 13 May 2024; Received in revised form 27 June 2024; Accepted 20 July 2024

Available online 29 July 2024

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Nomenclatures			
a_s	Outside surface solar absorptivity (-)	u	Flow velocity vector ($m.s^{-1}$)
C_{el}	Cost of electricity ($\$.kwh^{-1}$)	U	Overall thermal transmittance ($W.m^{-2}.K^{-1}$)
C_{enr}	Cost of energy consumption ($\$.m^{-2}$)	V_{Wind}	Wind velocity ($m.s^{-1}$)
C_{ins}	Cost of insulation material ($\$.m^{-3}$)	x,y	Cartesian coordinates (m)
C_p	Specific heat capacity ($kJ.kg^{-1}.K^{-1}$)	Greek symbols	
d	Discount rate (%)	$\nabla(-)$	Gradient operator
DD_c	Cooling degree-days ($^{\circ}Cdays$)	$\nabla.(-)$	Divergence operator
DD_h	Heating degree-days ($^{\circ}Cdays$)	β	Coefficient of thermal expansion by volume (K^{-1})
e_{ins}	Insulation thickness (m)	ρ	Density ($kg.m^{-3}$)
g	Acceleration due to gravity ($m.s^{-2}$)	σ	Stefan-Boltzmann ($W.m^2.K^4$)
h_e	Outdoor heat transfer coefficient ($W.m^{-2}.K^{-1}$)	λ	Thermal conductivity ($W.m^{-1}.K^{-1}$)
h_i	Indoor heat transfer coefficient ($W.m^{-2}.K^{-1}$)	η_s	Heating system efficiency (-)
H_u	Heating value ($J.m^{-3}$)	ε_i	Emissivity (-)
i	Inflation rate (%)	$\bar{\tau}$	Stress tensor (-)
K_{ins}	Thermal conductivity of insulation material ($W.m^{-1}.K^{-1}$)	Superscripts/subscripts	
N	Analysis period (<i>year</i>)	E	External
Q_c	Yearly cooling transmission loadc ($J.m^{-2}$)	I	Internal
Q_h	Yearly heating transmission load ($J.m^{-2}$)	Max/Min	Maximum/Minimum
q_i	Heat flux ($W.m^{-2}$)	Acronyms	
R_s	Solar radiation ($W.m^{-2}$)	Ak	Knitting acrylic
R_{wt}	Thermal resistance ($m^2.K.W^{-1}$)	As	Spinning acrylic
t	Time (<i>s</i>)	CFD	Computational Fluid Dynamics
T_a/T_e	Outside air temperature (<i>K</i>)	COP	Coefficient of performance
T_i	Indoor air temperature (<i>K</i>)	$LCCA$	Life Cycle Cost Analysis
T_{in}	Indoor surface temperature (<i>K</i>)	NPV	Present value factor
T_{max}	Maximum sol-air temperature (<i>K</i>)	OIT	Optimal insulation thickness
T_{min}	Minimum sol-air temperature (<i>K</i>)	UDF	User-defined function
T_{sa-UDF}	Sol-air temperature-UDF (<i>K</i>)	Wc	Carpet wool
$T_{sol-air}$	Solar air temperature (<i>K</i>)	Wr	Raw wool
T_{surr}	Average surface temperature (<i>K</i>)		

carbon footprint of construction projects and enhance energy efficiency, aligning with global efforts to mitigate climate change.

By incorporating these passive design strategies and cutting-edge insulation technologies, the construction industry in the Mediterranean can make significant strides towards sustainability. These efforts address immediate energy consumption challenges while contributing to long-term environmental stewardship and resilience against changing climate conditions.

In response to the challenges posed by high energy consumption and CO2 emissions in the construction industry, thermal insulation has become a focal point of research in recent years. Effective thermal insulation is crucial for reducing energy usage and enhancing the overall energy efficiency of buildings. Various innovative materials and techniques have been explored to improve the thermal performance of building envelopes.

One of the most promising developments in this field is the use of advanced insulating materials, such as aerogels, which offer exceptional thermal conductivity properties. Aerogels are highly porous and lightweight, making them an excellent choice for thermal insulation. However, their high initial cost and complex manufacturing processes present challenges that need to be addressed to make them more commercially viable.

In addition to aerogels, other innovative materials like expanded polystyrene (EPS), extruded polystyrene (XPS), polyurethane (PU), and glass fiber (GF) have been widely studied for their insulation properties (H.Huang et al., 2020). Each of these materials has its own set of advantages and limitations. For instance, EPS and XPS are known for their excellent thermal resistance and moisture resistance, while PU offers superior insulating properties and mechanical strength. Glass fiber is

valued for its high-temperature resistance and non-combustibility.

Researchers are also investigating the integration of composite materials with high thermal conductivity to enhance the effectiveness of traditional insulation materials. These composites aim to address common issues such as thermal bridging, which can significantly reduce the overall insulation performance of a building.

Thermal insulation, while essential for improving the energy efficiency of buildings, can have a very negative impact on the environment due to the embodied energy associated with the production of synthetic materials like polystyrene foams and polyurethanes. These materials require significant amounts of energy and resources for their manufacturing, greatly increasing their carbon footprint. In contrast, bio-based materials, such as those studied by Ouhaibi et al.(S.Ouhaibi et al.,2021; M. El Wazna et al., 2018; O. Mraggi et al., 2020; O. Mraggi et al., 2020), offer an eco-friendly alternative. These materials, derived from renewable sources like hemp, flax, and straw, are not only effective in terms of thermal performance but also have a significantly lower environmental footprint, thereby contributing to the environmental sustainability of construction projects.

Moreover, the concept of double-wall masonry structures has gained traction as an innovative construction approach to improve thermal insulation (F. Patania et al., 2010).This method involves the integration of internal air layers within the building envelope, creating a buffer zone that reduces heat transfer and enhances energy efficiency (T. Zhang et al., 2016). Depending on whether the air layer is open or closed, this configuration with two masonry wythes can significantly improve thermal performance by minimizing heat loss in winter and reducing heat gain in summer.

Local climate conditions and building usage patterns play a critical

role in selecting the appropriate insulation materials and construction methods. For instance, in regions with extreme temperature fluctuations, combining bio-based insulation materials with innovative construction techniques, like double-wall masonry structures, can provide superior energy efficiency and thermal comfort compared to traditional single-wall construction. This adaptability makes these methods applicable to a wide range of building types, from residential homes to commercial buildings.

In summary, addressing the high energy consumption and CO2 emissions in the construction industry requires a multifaceted approach focusing on bio-based thermal insulation materials and innovative construction techniques. By leveraging the unique properties of materials like aerogels and integrating them with construction methods such as double-wall masonry structures, significant strides can be made toward reducing energy consumption and enhancing the overall energy efficiency of buildings. This integrated approach aligns with global sustainability goals and highlights the importance of continuous research and development in the field of building energy efficiency.

The aim of this study is to determine the ideal thickness of sisal fiber, a sustainable thermal insulation material, for each of Morocco's six climatic zones. The main aim is to reduce energy consumption in summer and winter. To achieve this, we incorporate the degree-day concept and life-cycle cost analysis into our methodology. As shown by the majority of studies, notably (O.Kaynakli, 2012), life-cycle cost analysis has often been applied to calculate the optimum thickness of traditional insulation materials. Additionally, we assess the effectiveness of our novel air-gap solution and calculate energy requirements using computational fluid dynamics (CFD), a two-dimensional numerical fluid mechanics model. This technology is essential for improving energy efficiency in building design as it enables precise identification of optimal insulation thickness.

The study is organized into three main sections: the introduction, the body, and the conclusion. In the introduction, we provide an overview of the study's objectives and significance. Section II reviews the methodology and materials used in the investigation. Section III focuses on verifying our model, ensuring its accuracy and reliability. Finally, Section IV presents the study's findings, detailing the optimal insulation thickness for building walls in each climate zone. This structured approach allows us to thoroughly explore and understand the impact of sisal fiber insulation on energy efficiency across different climatic conditions in Morocco.

2. Materials and methods

2.1. Materials

The material used in this study is a nonwoven mat made of sisal fiber that is used as thermal insulation. A number of experimental tests and studies were used to assess this material's microstructure and thermo-physical characteristics (S. Ouhaibi et al., 2022). Standardized tensile tests were also used to systematically evaluate its tensile strength. The results show that the sisal fiber nonwoven mat sample's thermal conductivity coefficient, which was measured at 0.038 W/(m.K), is comparable to that of conventional insulation materials. This bio-based, environmentally friendly material has a great deal of promise for use as a thermal insulation solution in the building sector due to its low thermal conductivity and acceptable mechanical qualities.

In Table 1, ouhaibi et al. (S. Ouhaibi et al . 2022) outlined the thermo-physical properties of sisal fiber.

Table 1
The thermo-physical properties of the sisal fiber (S. Ouhaibi et al., 2022).

Materials	Thermal conductivity (W/m.K)	Specific heat (KJ/Kg.K)	Density (Kg/m³)
sisal fibre	0.038	1.245	134.83

2.2. Methods

2.2.1. Weather-related data

In our study, we analyze meteorological data collected over a decade from 37 monitoring stations spread across Morocco. These climatic zones are primarily characterized by the cumulative degree-days observed during both summer and winter. The resulting map, aimed at facilitating the enforcement of future regulatory measures, outlines six distinct climate zones. These zones are associated with specific cities, aligning with administrative borders (see Table 2):

2.3. Heat and cool demands associated for exterior walls

The assumption is made that heat loss exclusively takes place at the surfaces of external walls.

The yearly heating Q_h and cooling Q_c loads can be estimated utilizing the concept of degree-day, as outlined in Eqs. (1) and (2) (E. Küçüktopcu et al., 2018).

$$Q_h = 24 \times 3600 \times DD_h \times U$$
 (1)

$$Q_c = 24 \times 3600 \times DD_c \times U$$
 (2)

Where DD_h and DD_c represent the number of degree-days for heating and cooling needs respectively. The coefficient U expresses the overall thermal transmittance through the wall. It is calculated using (3):

$$U = 1/(R_{wt} + e_{ins}/K_{ins})$$
 (3)

Where R_{wt} represents the thermal resistance of the entire wall in the absence of insulating material, K_{ins} and e_{ins} represent respectively, the thermal conductivity and thickness of the insulating material.

2.4. The studied configuration

In Morocco, the envelopes of structures are usually built of brick and cement mortar. This study aims to evaluate the effect of sisal fiber integration on heating and cooling requirements and assess how it can enhance the thermal efficiency of outside walls. The first wall configuration, shown in "Fig. 1," is a composite wall with 20 cm of solid bricks and 2 cm of interior/exterior cement mortar. The second arrangement, which is also shown in "Fig. 1," has a similar architecture but adds a sisal fiber layer that is 2 cm thick that acts as insulation. Table 4 (M. Fellah et al., 2013) provides specifics on the materials' thermal characteristics.

2.5. Life cycle cost analysis (LCCA)

Thermal insulators must be incorporated into building envelopes in order to reduce energy use for both heating and cooling. The attainment of maximum efficiency is contingent upon precisely ascertaining the appropriate insulator thickness in relation to its characteristics. Throughout the building's existence, life cycle cost analysis is a proven

Table 2
Description of the studied climate (Ameé, RTCM, 2015).

Climate zone	Cities	Climate type	Climate characteristics
1	Agadir	Oceanic	Mild, humid winters & Summers tempered by the ocean
2	Tangier	Mediterranean	Mild, rainy winters & Hot, dry summers
3	Fez	Semi-arid	Cold winters with frequent frost Very hot & Dry summers
4	Ifrane	Mountain	Cold, snowy winters & Mild summers
5	Marrakech	Semi-arid	Mild winters & Very hot, dry summers
6	Errachidia	Arid	Cool winters & Torrid summer

Table 3 DD_h and DD_c of chosen cities (A. Benallel et al., 2022).

City	Latitude (°)	Altitude (m)	Longitude (°)	DD_h (° C-day)	DD_c (° C-day)
Agadir	30.4	23	−9.6	619	549
Tangier	35.7	85	−5.8	830	441
Fez	33.9	579	−5.0	1295	712
Ifran	33.5	1665	−5.2	1901	517
Marrakech	31.6	466	−8.0	754	1042
Errachidia	31.9	1045	−4.4	1034	1407

method for reducing the discounted total energy cost. This approach takes into account variables like the discount rate (d), the inflation rate (i), and the energy usage related to the thickness of the insulator. It is easier to estimate the total cost per unit of wall surface when using Eq. (4) (M. Dlimi et al., 2019).

$$LCC = \underbrace{C_{ins} \times e_{ins}}_{C_i} + \underbrace{NPV \times C_{enr}}_{C_e} \quad (4)$$

NPV represents the present value factor, C_{enr} the total annual energy cost, C_{ins} represents the cost per unit volume of the thermal insulator, and e_{ins} the thickness of this insulator.

The annual energy cost (\$/m²) constitutes the total of the yearly heating and cooling expenditures, determined through Eq. (5) (M. Dlimi et al., 2019):

$$C_{enr} = \frac{Q_c \times C_{el}}{COP \times 3.6 \times 10^6} + \frac{Q_h \times C_{el}}{H_u \times \eta_s} \quad (5)$$

COP represents the coefficient of performance of the cooling system. C_{el} is the cost of electricity. H_u corresponds to the lower heating value of electricity. Finally, η_s is the efficiency of the heating system.

NPV was calculated taking into account Discount rate (d) and the inflation rate (i), using the following expressions:

$$NPV = \frac{(1+r)^N - 1}{r(1+r)^N} \text{ if } i > d; \quad r = (i-d)/(i+d) \text{ or } i < d; \quad r = (d-i)/(i+1)$$

$$NPV = \frac{N}{1+i} \text{ if } i = d$$

- Constraints and Limitations of the LCCA

The calculation method used for the life-cycle cost analysis (LCCA) has several drawbacks, including the following:

- Limited Scope of Insulation Thickness Optimization: The optimum insulation thickness was determined exclusively based on a single multi-layer wall and specific climates.
- Exclusion of Embodied Energy: In calculating the energy cost, only the energy related to the operational phase (operational energy) was considered, while the energy related to the production phase (embodied energy) was not taken into account.
- Narrow Cost Criterion: The cost of insulation was determined solely based on the price of the insulating material, without accounting for additional costs associated with installing and operating the insulation.

2.6. Simulation model

In this section, we will evaluate through numerical simulation the thermal performance and dynamic behavior of two wall configurations:

- The first one with sisal fiber insulation material

Table 4

Construction materials and their thermophysical characteristics (M. Fellah et al., 2023).

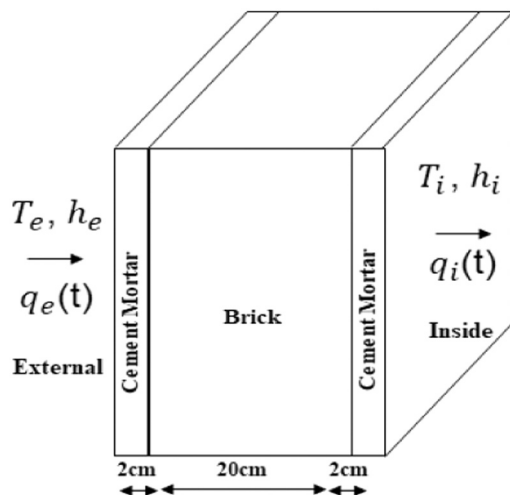
Materials	Cement Mortar	Brick	Insulation
Thickness (cm)	2	20	2
Thermal conductivity (W/m.K)	0.97	0.925	0.038
Specific heat (KJ/Kg.K)	0.84	0.92	1.245
Density (Kg/m ³)	1600	2100	134.83

Table 5

Factors considered in economic assessment.

Parameter	Value
Electricity cost (C_{el}), (ONEE, 2014)	0.10 \$/KWh
Coefficient of performance (COP), (Gounni et al., 2019)	3
Heating value (H_u), (Bolattürk, 2006)	3.6×10^6 J/m ³
Efficiency (η_s), (Bolattürk, 2006)	0.99
Insulation cost (C_{ins}), (Ouhaibi et al., 2022)	30.43 \$/m ³
Inflation rate (i), (Benallel et al., 2022)	2.5 %
Discount rate (d), (Benallel et al., 2022)	5.5 %
Period of analysis (N), (Dlimi et al., 2019)	20 years

Uninsulated configuration



Insulated configuration

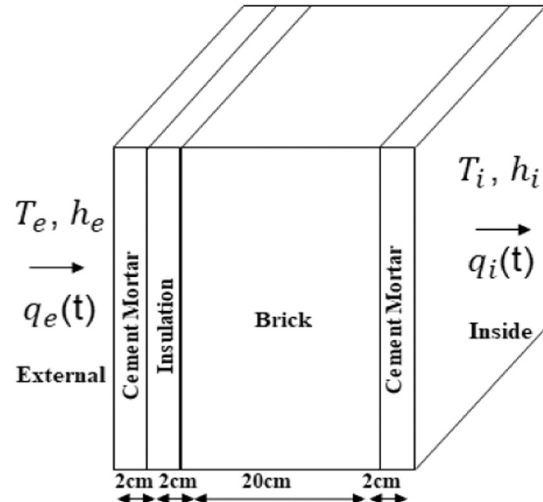


Fig. 1. Layout of the analyzed wall.

- The second one with an intelligent solution integrating an air gap to minimize the required insulation thickness

The simulation will utilize the CFD method using Ansys/Fluent 19.0 software. We will apply this dynamic analysis to a south-facing wall located in the Ifrane climate zone, one of the highest energy-consuming areas in Morocco.

We shall elucidate the physical model utilized for simulating heat transfer within the analyzed wall. Conduction takes place within the solid materials of the wall: cement mortar, bricks and sisal fiber. Convection occurs in the fluid part: air gap, and between the wall and the inside.

The assumptions made for the CFD computations are as follows:

- Laminar air flow with unsteady heat transfer
- 2D modeling
- Combined conduction-convection transfers at the solid-fluid interface
- Isotropic properties of materials
- Convection within the air-filled gap is assumed to be natural ($Ra < 10^6$)
- Air is considered as a Newtonian fluid, and it is assumed that the shear stress resulting from the sliding of air at the wall interfaces of the gap is negligible.

2.6.1. The equation for energy

The diffusion/convection equation is solved to describe the dynamic evolution of air temperature fields in the cavity and the heat crossing the different sections, as indicated in (6) (O. Iken et al., 2019). Where C_p is the specific heat and λ is the thermal conductivity.

$$\rho C_p \frac{\partial T}{\partial t} + \rho C_p u \cdot \nabla T = \nabla \cdot (\lambda \nabla T) \quad (6)$$

2.6.2. The equation for mass conservation

The resolution of the Navier-Stokes equations makes it possible to determine the instantaneous evolution of the velocity and pressure fields in the cavity (O. Iken et al., 2019), initially filled with air and of variable thickness depending on the case studied.

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \rho u = 0 \quad (7)$$

2.6.3. The equation for momentum conservation

The motion equations of the air filling the air gap are expressed by (8) (O. Iken et al., 2019).

$$\rho \frac{\partial u}{\partial t} + \left(\rho u \cdot \nabla \right) u = -\nabla p + \nabla \cdot \bar{\tau} - \rho \beta (T - T_{ref}) g \quad (8)$$

$$\bar{\tau} = \mu [(\nabla u + \nabla u^T)] \quad (9)$$

It should be noted that u is the flow velocity vector, p is the static pressure and $\bar{\tau}$ is the stress tensor described in (9). In addition, the term $\rho \beta (T - T_{ref}) g$ designates the gravitational body forces expressed according to the Boussinesq approximation (O. Iken et al., 2019).

- Frontier conditions

Eqs. (10) and (11) delineate the boundary conditions for the outer and inner wall surfaces, correspondingly. The remaining facets of the wall are presumed to be adiabatic, as denoted in Eq. (12).

$$-\lambda_0 \left(\frac{\partial T}{\partial x} \right)_{x=0} = h_i (T_{sol-air}(t) - T_{x=0}) \quad (10)$$

$$-\lambda_l \left(\frac{\partial T}{\partial x} \right)_{x=L} = h_i (T_{x=L}(t) - T_{in}(t)) \quad (11)$$

$$\frac{\partial T(x, 0, t)}{\partial y} = \frac{\partial T(x, l, t)}{\partial y} = 0 \quad (12)$$

To calculate the combined radiative and convective heat transfer coefficients on the outer wall surfaces h_e , we employ Eq. (5).

$$h_e = D + EV_{wind} + FV_{wind}^2 \quad (13)$$

The empirical constants D , E and F depend on the surface type. V is the wind speed. For brick and plaster: $D = 12.49$, $E = 4.065$ and $F = 0.028$ (M. Mirsadeghi et al., 2013).

The indoor convective heat transfer coefficient h_i is determined by Eq. (14):

$$h_i = 3.08 + 5.72 \varepsilon_i \quad (14)$$

In this study, an emissivity ε_i of 0.90 is considered for the non-reflective surface. The indoor convective heat transfer coefficient h_i is estimated at $8.22 \text{ W/m}^2 \cdot \text{K}$.

$T_{sol-air}$ represents solar air temperature, which incorporates both solar radiation and outdoor temperature effects, is determined using Eq. (15), as proposed by Yumrutaş and al (R. Yumrutaş et al., 2007)

$$T_{sa}(t) = T_a(t) + \frac{a_s R_s(t)}{h_e} - \frac{\varepsilon \sigma (T_a^4 - T_{surr}^4)}{h_e} \quad (15)$$

Eq. (15) allows calculating the solar air temperature T_{sa} , which combines the effects of solar radiation R_s and ambient temperature T_a . This equation involves the solar absorptivity a_s , the combined convective-radiative heat transfer coefficient h_e , the emissivity ε , the Stefan-Boltzmann constant σ and the average surface temperature T_{surr} (J. Mohammed et al., 2004). The last term is generally negligible for vertical walls according to Kontoleon et al. (K. J. Kontoleon et al., 2008), due to the low effective sky temperature.

The Meteorom database (METEONORM V8.1.4, 2022) for the Moroccan city of Ifrane provides the meteorological information that was used. It is related to the two warmest (July 21–22) and two coolest (January 12–13) days of the year. This data includes solar radiation R_s , temperature T_a and wind speed for a south-facing orientation.

The sol-air temperature T_{sa} is first calculated with Matlab for a standard exterior surface ($a_s = 0.7$) according to (J. A. Reagan et al., 1979). Then T_{sa} is ascertained for the analyzed wall surface “Fig. 2”.

The 4 most extreme days (2 hottest and 2 coldest) were chosen in order to analyze the performance of the wall with sisal fiber and air gap. The Frontier conditions outlined in this study are specified in “Fig. 2”. The outdoor surface temperature is assumed to be equivalent to the solar air temperature T_{sa} . This is analogous to a sinusoidal signal (16) “Fig. 3” and implemented in the CFD code via a user-defined function (D. Risberg et al., 2016).

$$T_{sa-UDF}(t) = \frac{T_{max} - T_{min}}{2} \sin\left(\frac{2\pi t}{24} - \frac{\pi}{2}\right) + \left(\frac{T_{max} + T_{min}}{2}\right) \quad (16)$$

T_{max} and T_{min} represent the maximum and minimum sol-air temperatures. In summer, they are 77.33°C and 23.20°C . In winter, they are 23.87°C and -3.3°C .

A one-dimensional heat transfer is considered from the outside (left) to the inside (right).

2.7. Measure of key response (MKR) index

Thermal performance of the wall with ecological insulation in the first configuration, and of the wall combining ecological insulation and air gap in the second configuration (see Fig. 2) is primarily determined by key parameters such as wall temperature variation, time lag, decrement factor, and wall heat gain. The time lag and decrement factor are

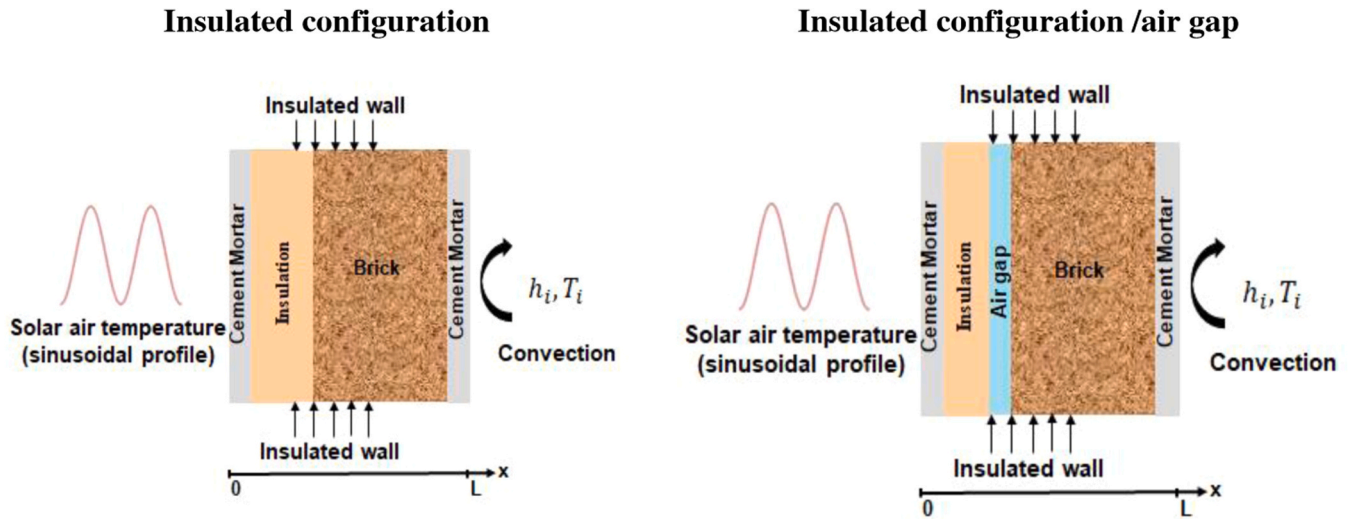


Fig. 2. The considered Boundary conditions.

critical for determining the efficiency of the arrangement in dampening the diurnal temperature cycle propagating through the wall. The time lag is the amount of time it takes for diurnal temperature waves to travel from the outside to the inside of the wall. In contrast, the decrement factor is the ratio by which the amplitude of the diurnal temperature wave is lowered between the wall's outer and inner surfaces.

For the best thermal performance, the largest time lag and lowest decrement factor are suggested. The time lag (Φ) and decrement factor (f) are determined using the formulae (17, 18) (M. Fellah et al., 2024).

$$f = \frac{T_{x=L, \max} - T_{x=L, \min}}{T_{x=0, \max} - T_{x=0, \min}} \quad (17)$$

$$\Phi = t_{T_{x=L, \max}} - t_{T_{x=0, \max}} \quad (18)$$

The heat gain of the envelope is also important in selecting the optimal arrangement for thermal performance. To achieve optimal thermal performance, the building must minimize heat gain, which is computed as the heat flux entering or exiting the wall. This study offers the Measurement of Key Response (MKR) index, which combines performance metrics such as time lag, decrement factor, and envelope heat gain. This index allows for a comparison of insulated and uninsulated walls. The MKR index is denoted as (19), (D. Bhamare, et al., 2020)

$$MKR = \frac{\Phi'}{(f' + c) \times q'} \quad (19)$$

Where Φ' , f' and q' denote the non-dimensional time lag, non-dimensional decrement factor, and non-dimensional heat gain or flow, respectively. In the case of an ideal wall with constant outside temperature, the non-dimensional decrement factor equals zero. To avoid division by zero in Eq. (19) for an ideal wall, a 0.5-valued computation constant c is introduced.

The aforementioned parameters are determined as follows.

$$\Phi' = \frac{\Phi_{\text{with-insulation}}}{\Phi_{\text{without-insulation}}} \quad (20)$$

$$f' = \frac{f_{\text{with-insulation}}}{f_{\text{without-insulation}}} \quad (21)$$

$$q' = \frac{q_{\text{with-insulation}}}{q_{\text{without-insulation}}} \quad (22)$$

3. Validation model

3.1. Mesh sensitivity study

Exploring mesh sensitivity is a pivotal stage in numerical analysis, as it increases the accuracy and relevance of the results obtained. Certainly, the latter should remain unaltered by spatial discretization.

Five mesh grids were proposed for the sensitivity analysis: G_1 with 1122 elements, G_2 with 3596 elements, G_3 with 92070 elements, G_4 with 112832 elements and G_5 with 142396 elements. Spatial discretization involved the utilization of mesh grids composed of quadrilateral elements (Fig. 4). A performance metric was devised, tracking the evolution of inner surface temperature over time, facilitating the assessment and comparison of each grid's influence on this metric (Fig. 5). Through this method, the number of elements at which the alteration in interior temperature remains below a convergence threshold specified by the grid can be determined.

Noticing the convergence of the temperature variation among the last three grids, as depicted in Fig. 3, we designate grid as the optimal number of elements. The results demonstrate swift convergence and maintain independence from spatial discretization. Moreover, a second-order numerical scheme has been employed for both the discretization of dynamics and energy interpolation.

It is worth noting that a constant time step was employed to replicate heat transfer mechanisms via conduction and convection in the air gap. The numerical technique used is iterative with relaxation parameters set at 0.75 for energy and 0.5 for momentum. This approach minimizes computation time while preserving the significance of the results. The "Simplec" resolution approach was utilized to solve the pressure-velocity coupling using an implicit strategy. Additionally, a second-order numerical method was applied to discretize the momentum and energy equations, as well as the pressure interpolation method. Finally, the numerical findings were confirmed to be convergent when the air velocity and energy residuals were less than 10^{-6} .

3.2. Benchmarking experience

The numerical results are compared with the experimental data of El wazna et al. (M. El Wazna et al., 2018) in order to evaluate the reliability of the numerical methods employed in this investigation, as well as the validity and correctness of the model used. Based on four different types of waste raw wool, carpet wool, spinning, and knitting acrylic referred to here as W_r , W_c , A_s , and A_k , respectively this study examines textile insulation in the form of a non-woven fabric. The following steps are

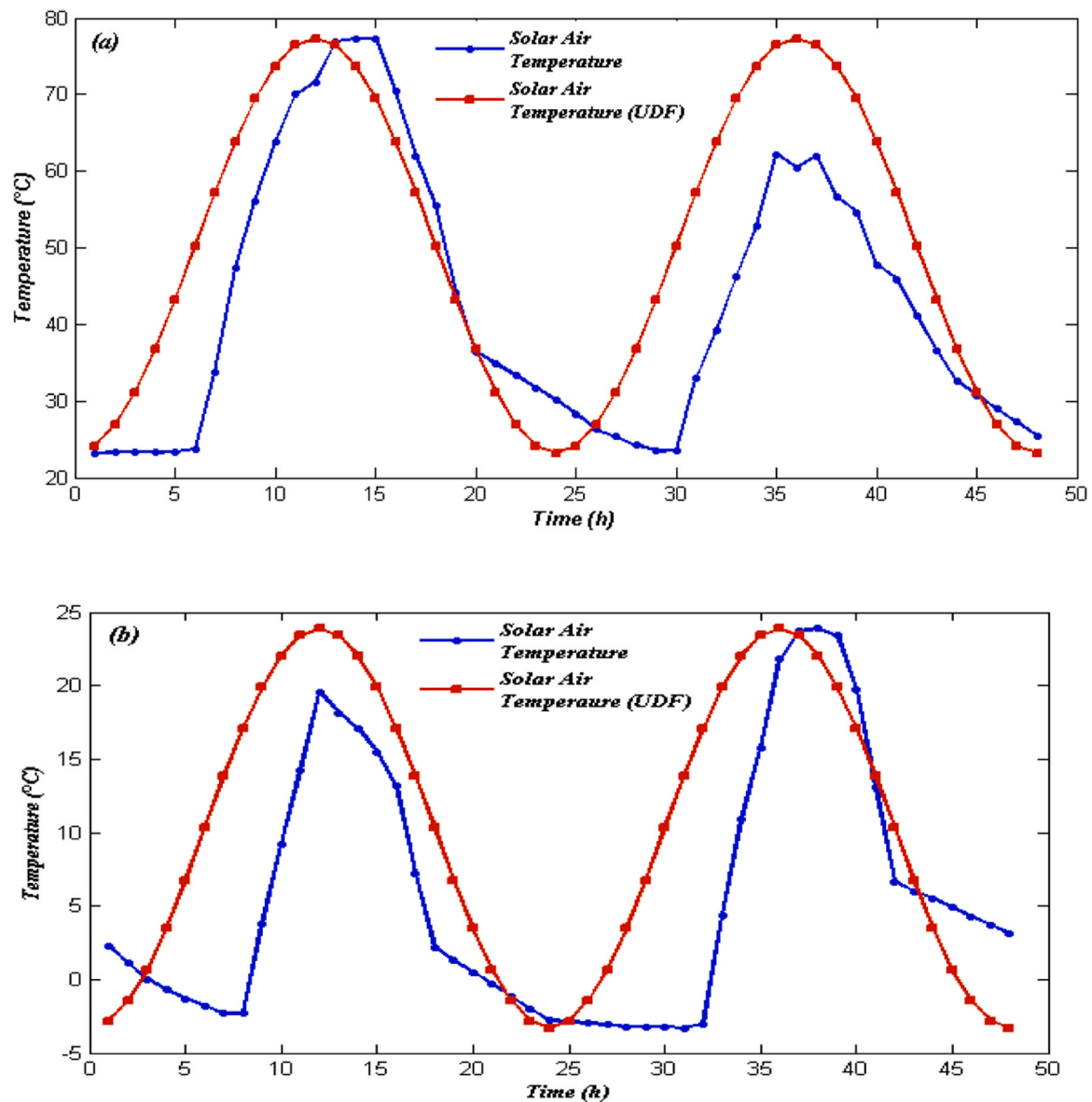


Fig. 3. External weather conditions: (a) summer solar-air temperature, (b) winter solar-air temperature.

involved in the production of non-woven insulation: gathering waste textiles, shredding, creating staple fibers, opening, cleaning, and blending, carding, laying webs, and mechanical bonding, which results in the non-woven material. Table 6 (M. El Wazna et al., 2018) displays the thermo-physical characteristics of textile insulation.

The investigations are conducted in a thermoregulated cavity, a reduced-scale test cell. This test cell is shaped like a cubic box and has 0.4 m x 0.4 m x 0.4 m wooden walls that are removable and interchangeable (see Fig. 4) [22]. The box's floor and ceiling are composed of a highly insulating material. An incandescent bulb set inside a black window cover and placed in the test cell's middle provides heat for the device. Samples of textile insulation are sandwiched between two wood layers that are one centimeter thick.

The temperature's time progression is seen in Fig. 5. Although the agreement is not as good as first anticipated, it is nevertheless considered satisfactory, with relative errors varying from 0.1 % to 10 % at various time intervals. The observed disparities can be ascribed to intrinsic uncertainties in experimental measurements, arising from the combination of the accuracy constraints of measuring devices and any fluctuations in the test cavity conditions.

4. Results and discussion

4.1. Heating and cooling load's analysis

In winter, the uninsulated configurations exhibit heating energy demands of approximately 34.71; 46.55; 72.63; 106.62; 42.28; and 57.99 kWh/m² for Agadir, Tangier, Fez, Ifrane, Marrakech, and Errachidia, respectively. With insulation, these values decrease to 15.56; 20.87; 32.57; 47.81; 18.96; and 26.01 kWh/m², representing a notable reduction in demand.

During summer, the uninsulated configurations show consumptions of 30.8; 27.73; 39.93; 28.99; 58.44; and 78.91 kWh/m². With insulation, these values decrease to 13.8; 11.09; 17.90; 13.01; 26.20; and 35.38 kWh/m² for the corresponding climatic zones.

These data highlight the energy demands for heating in winter and energy consumption in summer across various climatic zones in Morocco. The integration of insulation leads to a significant reduction in these figures, underscoring the pivotal role insulation plays in diminishing energy consumption for both heating in winter and cooling in summer. Such reductions can substantially enhance the energy

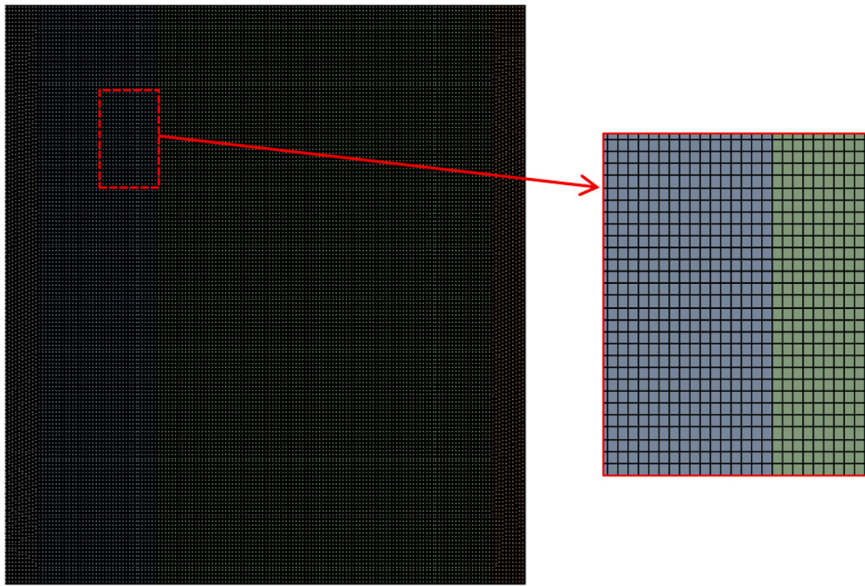


Fig. 4. The mesh grid of the studied configuration.

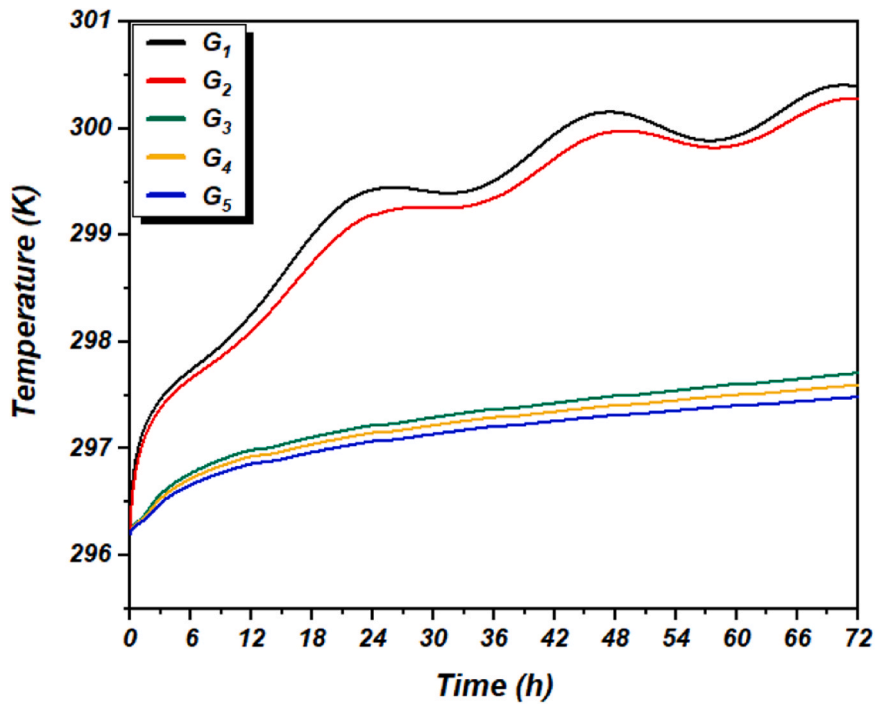


Fig. 5. Mesh sensitivity analysis results.

efficiency of buildings and yield considerable energy savings. It's imperative to recognize that lower energy consumption not only reduces energy costs but also mitigates the carbon footprint associated with building heating and air conditioning systems.

4.2. Optimum insulation thickness determination via life cycle cost analysis

The calculated thermal loads for each of the six Moroccan climate zones are used to calculate the optimum insulation thickness (OIT). The life cycle cost analysis (LCC) uses these thermal loads as inputs.

The evolution of insulation, energy, and overall costs as a function of insulation thickness is shown in Fig. 7. Although adding more insulation

results in a linear increase in insulation prices, it lowers energy costs.

The whole insulation and energy expenses are combined to create the OIT, which is the lowest possible total cost. The following OIT values were obtained: 0.04 m for Agadir; 0.04 m for Tangier; 0.06 m for Fez; 0.07 m for Ifrane; 0.05 m for Marrakech; and 0.06 m for Errachidia.

The calculation of the ideal insulation thickness (OIT) in Morocco's several climate zones is explained in the text, taking into account thermal loads. The impact of insulation thickness on energy and insulation costs is depicted in the figure. When accounting for both variables, the thickness with the lowest overall cost is called the OIT. OIT values vary throughout cities, which is indicative of their distinct climates and energy expenses.



Fig. 6. Experimental test system.

Table 6
Thermo-physical properties of textile insulation (M. El Wazna et al. 2018).

Textile insulation	Thickness (cm)	Thermal conductivity (W/m.K)	Specific heat (KJ/Kg.K)	Density (Kg/m ³)
Wr	1.3	0.03476	1.200	8.153
Wc	1.2	0.03827	1.200	10.583
As	1.4	0.0429	1.433	14.571
Ak	1.2	0.0431	1.433	11.33

4.3. Simulation results

4.3.1. Air velocity analysis

Air velocity is an essential graphical indicator of performance. It characterizes the quantity of air movement within the air gap, and is directly associated with convective heat transfer. As shown in Fig. 8, the variation in air velocity for the configuration studied during the warm and cold seasons, at two distinct times: 16:00 (t = 40 h) and 00:00 (t = 48 h) of the second day.

The air gap analyzed is closed, meaning that air movement inside it is governed exclusively by the ground-air temperature. During the warm

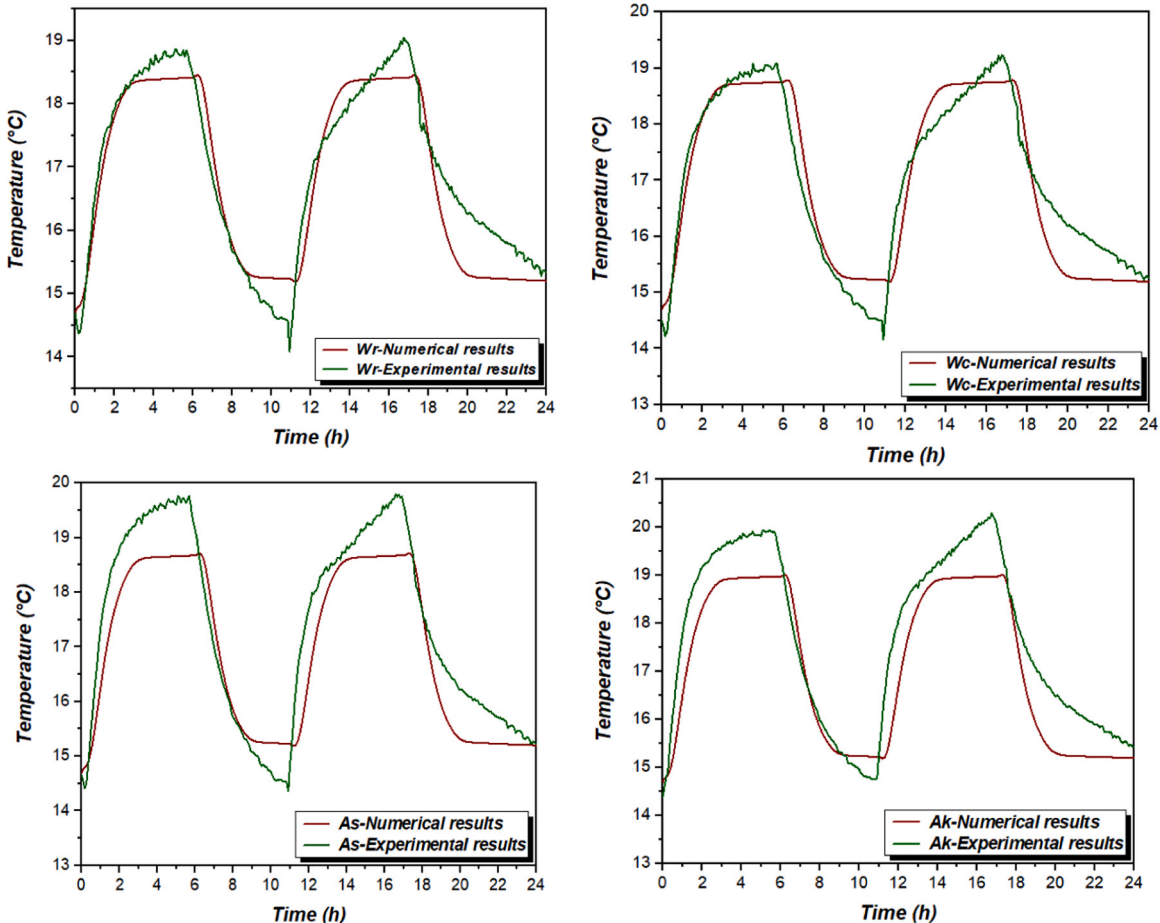


Fig. 7. Comparison between numerical results and experimental data. (M. El Wazna et al., 2018).

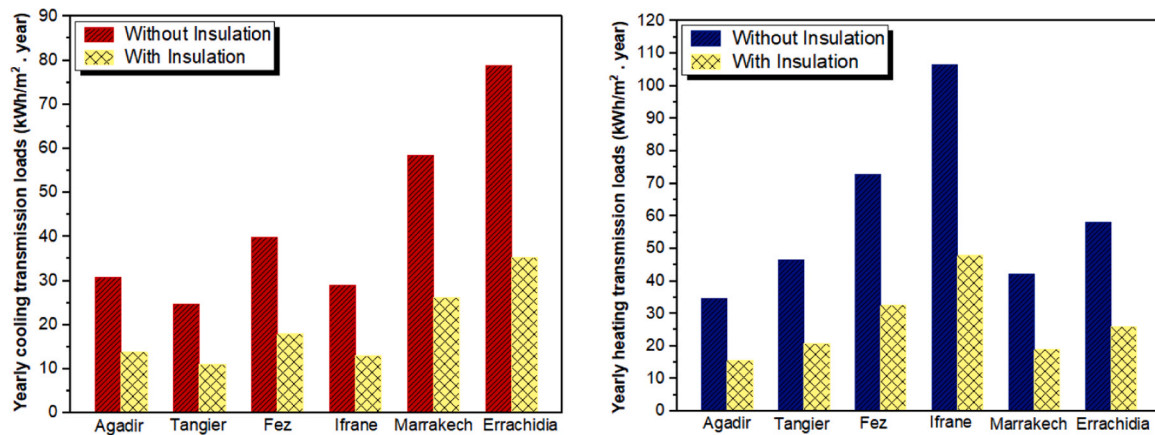


Fig. 8. Transmission loads.

season, the ground-air temperature generally remains above 25°C. Under these conditions, the air in the blade receives more solar heat, resulting in a maximum velocity of around 0.058 m/s, compared to the cold season when the maximum velocity is 0.037 m/s. The air velocities shown in Fig. 8 detail the velocity distribution inside the air knife. It is noticeable that the movement of the air mass is particularly concentrated near the inner surfaces of the façade. On the other hand, at the bottom and top of the blade, air mass movement is considerably weaker. This is attributed to the high heat transfer near the thermal insulation.

4.3.2. Examination of temperature contours

In this section, we present and examine the temperature contours for two distinct configurations: that with the sisal fiber insulation and that with the insulation associated with the air gap. These contours are studied in both summer and winter, at two different times of day ($t=40$ h and $t=48$ h). The aim is to assess the impact of the insulation configuration and the intelligent air gap solution on the temperature of the wall's interior surface. These contours detail the propagation of the thermal wave through the various wall components.

The first configuration with insulation includes solid materials such as cement plaster and bricks, in addition to sisal fiber insulation. In this way, the shape of the isotherms is a straight line normal to the direction of heat transfer. In summer, inner wall surface temperatures are around 23°C for both times (at $t=40$ h and $t=48$ h), as shown in Fig. 9. These values are close to the comfort temperature of 26°C. In winter, the temperatures of the interior cement plaster layer do not exceed 5°C to 7°C at times $t=40$ h and $t=48$ h, as shown in Fig. 9. These values are considerably lower than the winter comfort reference of 20°C. Thus, this configuration results in high heating energy consumption, as shown in Fig. 10.

Fig. 9 shows the temperature contours in the ingenious 2 cm air gap configuration. Due to the reduced thickness of the air gap, there is a slight inclination with a subtle undulation in the shape of the isotherms. Consequently, the impact of this 2 cm air gap on the heat wave is limited. This observation is corroborated by temperature values on the inside of the wall. In summer, temperatures vary between 23.5°C and 23.2°C on the inner layer of cement plaster at times $t=40$ h and $t=48$ h of the day, as shown in Fig. 9. These surface temperature values on the inside are closer to the comfort temperature of 26°C than in the first configuration, but remain insufficient due to the limited thickness of the air space.

Similarly, in winter, there is a slight increase in the temperature of the inner wall surface compared to the configuration insulated with sisal fiber. In fact, in this configuration, the temperature of the interior surface increases by around 1°C compared with the wall of the first sisal-fiber-insulated configuration, at both instants. The low air velocity values shown in Fig. 8 also confirm the moderate impact of this

configuration, although it is still better than the first configuration.

4.3.3. Evaluation of heating and cooling demands

In Morocco, prevalent building materials include brick, concrete, and cement mortar. The outdoor climate often subjects many structures to uncomfortable thermal conditions. The objectives of this study are to analyze the insulating effect of sisal fiber and determine its optimal thickness to reduce thermal loads. Following the significant insulation need identified, especially in the Ifrane area, we propose an intelligent solution: adding a 2 cm unventilated air gap to minimize the sisal thickness to 5 cm. This constant air gap also allows solar radiation to pass through in winter, becoming thermal insulation.

The instantaneous heat transfer method is used to calculate the daily heating and cooling loads, for the wall configuration with 7 cm of sisal fiber and the 2nd configuration with 5 cm of sisal fiber and 2 cm air gap. This method is based on the heat flux q_i exchanged between the inner wall surface and the indoor environment.

$$q_i = h_i(T_{x=L} - T_{in}) \quad (23)$$

Eq. (23) is used to calculate the thermal loads. It takes three inputs: the inner surface temperature T_{in} , the indoor convective heat transfer coefficient h_i , and the setpoint temperature (20°C in winter and 26°C in summer). "Fig. 10" shows the total daily loads, which are obtained by integrating over a 24-hour period.

The simulations reveal that during the winter, the configuration with 7 cm of sisal fiber insulation results in a high heating demand of 2.51 kWh/m². However, the second configuration with an air gap significantly reduces this demand to 2.42 kWh/m².

In the summer, the first configuration exhibits similarly high air conditioning consumption, totaling 0.29 kWh/m². The presence of an air gap substantially decreases this to 0.28 kWh/m².

These findings demonstrate significant energy savings achieved by using an air gap, both for heating and cooling requirements, particularly in the Ifrane climate zone, which necessitated very high insulation thickness.

The text discusses the results of simulations related to heating and cooling energy demands for two different insulation configurations in the Ifrane climate zone. It highlights the significant reduction in energy demand when an air gap is introduced. The use of an air gap is particularly effective in areas with extreme climate conditions, such as Ifrane.

4.3.4. MKR index analysis

The MKR index can also be used to assess the thermal performance of the second configuration, which combines 5 cm of eco-friendly insulation and 2 cm of air space, compared with the first configuration, which includes only 7 cm of insulation. In summer, the configuration with 7 cm of sisal fiber insulation results in an MKR of 0.0833. However,

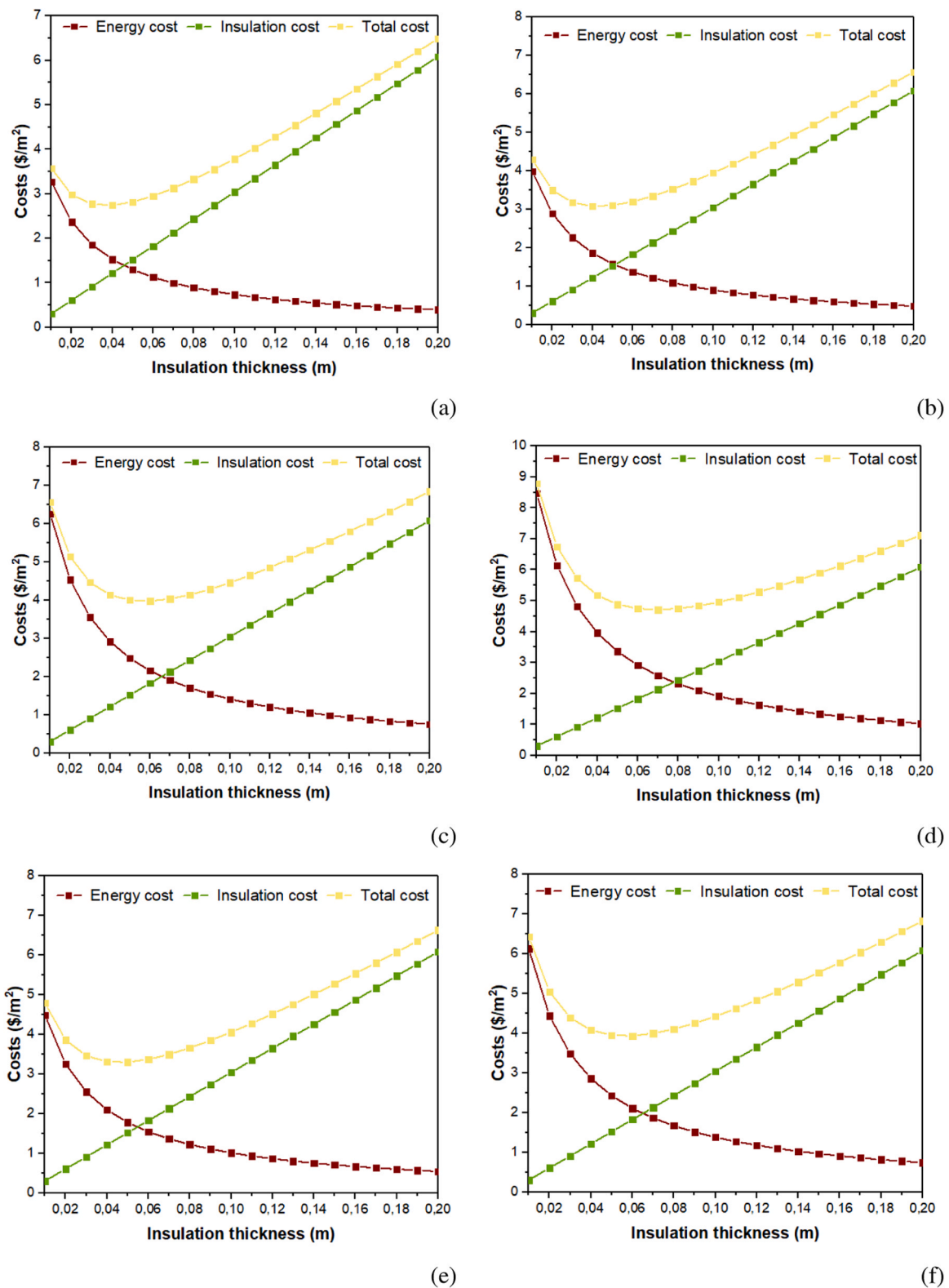


Fig. 9. Evolution of insulation, energy, and total costs versus insulator thickness, for the six Moroccan climate zones: (a)-agadir (b)-tangier (c)-fez (d)-ifrane (e)-marrakech (f)-errachidia.

when using the second configuration with a 2 cm air gap and a reduced thickness of 5 cm of insulation, the MKR index slightly increases to 0.0853. This slight increase suggests that the presence of the air gap marginally enhances the thermal performance of the wall, maintaining insulating efficiency while adding an additional air barrier.

In winter, the differences are more pronounced. The first configuration, with just 7 cm of insulation, has an MKR value of 0.7422. By introducing a 2 cm air gap in the second configuration, the MKR increases to 0.7481. This increase indicates that the air gap plays a more significant role in reducing heat transfer during the winter months, thus

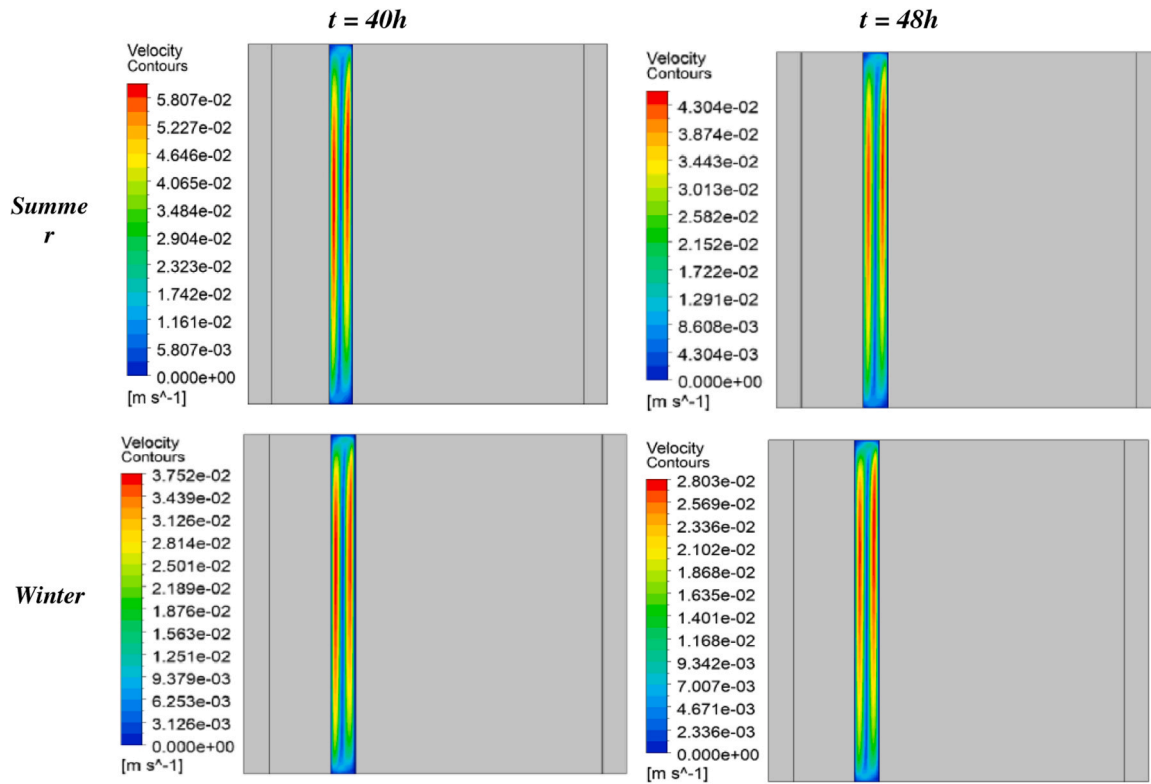


Fig. 10. Velocity contours.

contributing to better thermal insulation. The presence of the air gap creates an additional barrier against heat, improving the overall efficiency of the insulation system.

These results show that, although the addition of an air gap improves thermal performance in both seasons, the impact is more noticeable in summer than in winter. The air gap acts as a thermal buffer, reducing the heat passing through the wall and thereby enhancing thermal comfort inside the building. It also allows for a reduction in insulation thickness from 7 cm to 5 cm, as demonstrated in this case. This highlights the importance of designing insulation systems that incorporate air layers to maximize energy efficiency year-round. In conclusion, the MKR index demonstrates the effectiveness of the second configuration in Morocco's most energy-intensive climate zone, Ifrane, delivering significant improvements over simple insulation.

5. Conclusion

In this study, we harnessed the degree-day concept and conducted a comprehensive life cycle cost analysis to assess the thermal and economic performance of insulation materials within the walls of buildings across Morocco's diverse climate zones. By employing dynamic simulations, we were able to fine-tune the optimum thickness.

One of the key findings of this research is that the Optimal Insulation Thickness (OIT) varies with the magnitude of cooling and heating loads. For instance, in the Ifrane zone, where cooling and heating demands are substantial, the optimal insulation thickness is 7 cm. However, by implementing a smart solution that incorporates an air gap, we managed

to reduce this thickness to 5 cm. Notably, this adjustment resulted in an energy gain of 3.72 % during summer and 3.68 % during winter. These results are validated by the MKR index, which compares the two configurations studied.

These findings underscore the significance of tailored insulation strategies for different climatic zones and highlight the potential for innovative solutions, such as air gaps, to enhance energy efficiency in building design. Ultimately, this research serves as a valuable resource for optimizing building insulation and reducing energy consumption in various climates.

CRediT authorship contribution statement

Naoual Belouaggadia: Supervision. **Khalifa Mansouri:** Supervision. **Nassim SEBAIBI:** Supervision. **Salma Ouhaibi:** Validation, Methodology. **Mohammed FELLAH:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

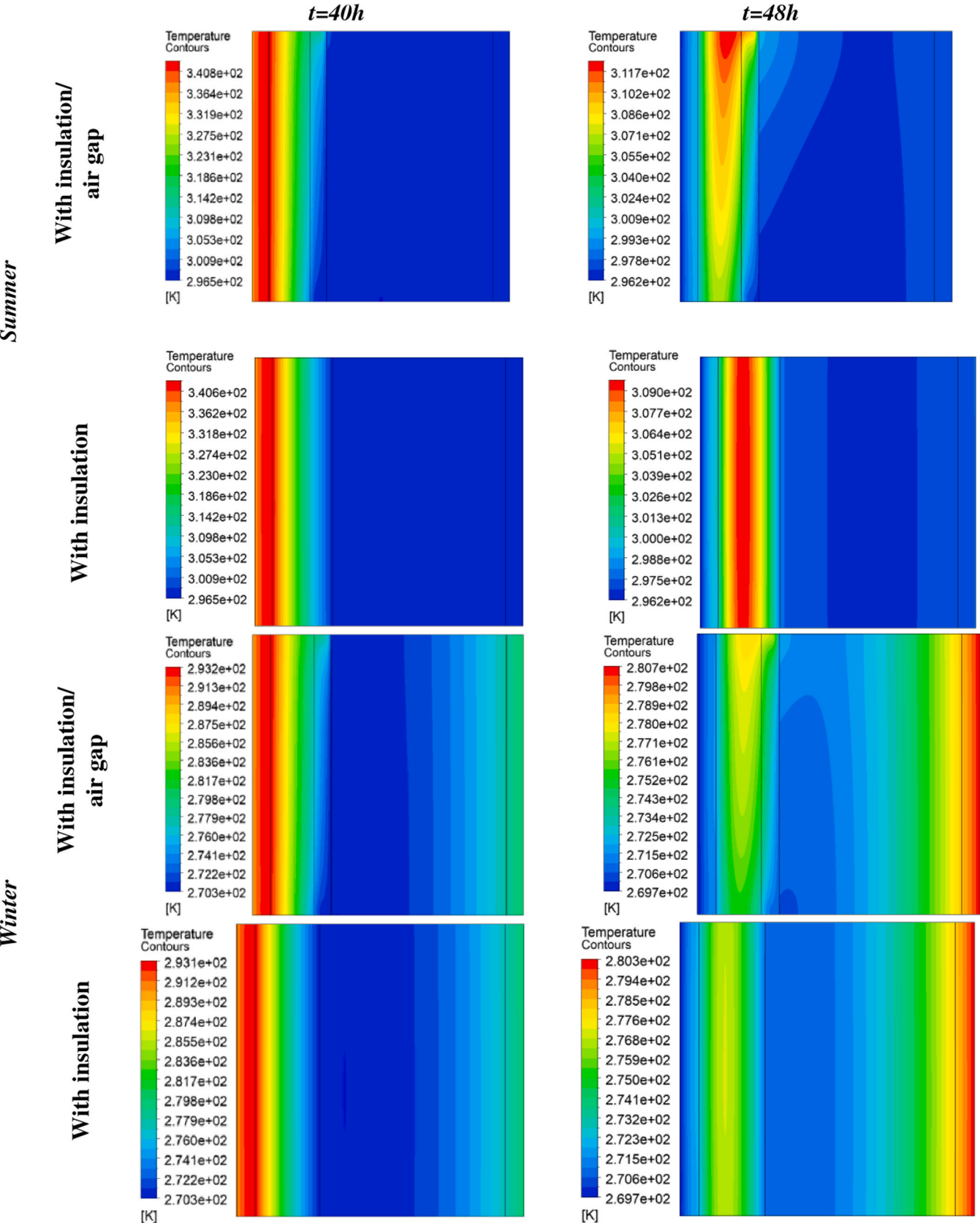


Fig. 11. Temperature contours.

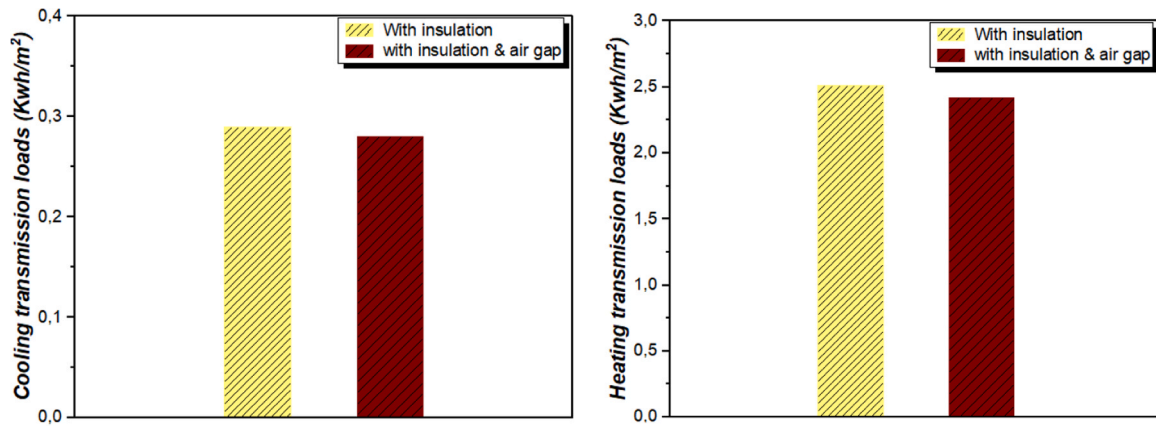


Fig. 12. Transmission loads.

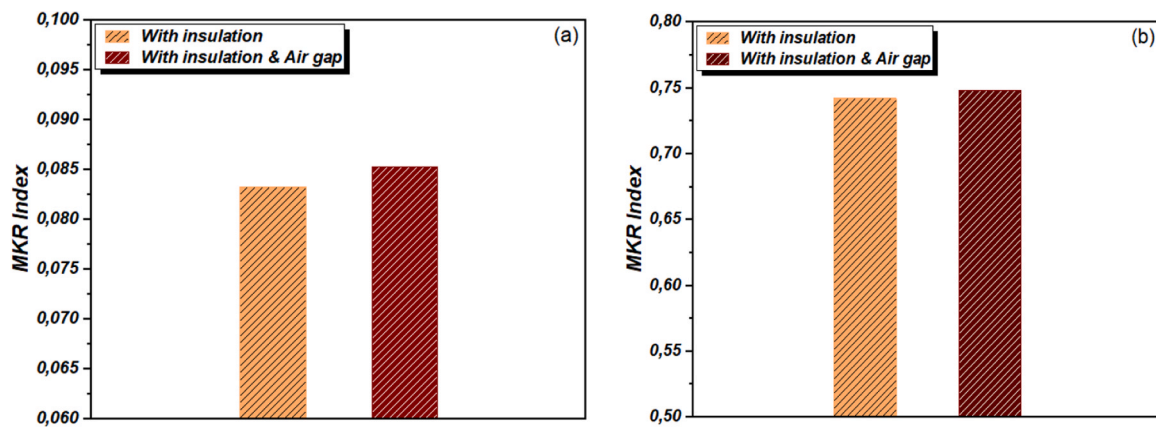


Fig. 13. MKR Index (a) summer day, (b) winter day.

Appendix. UDF code used in the CFD simulations

Code for validation

```
#include "udf.h"
DEFINE_PROFILE(temperature_profile, thread, position)
{
    face_t f;
    real time = CURRENT_TIME;
    real temperature = 288;
    if (time >= 0.0 & time < 18000.0)
    {
        temperature = 309;
    }
    else if (time >= 18000.0 & time < 36000.0)
    {
        temperature = 288;
    }
    else if (time >= 36000.0 & time < 54000.0)
    {
        temperature = 309;
    }
    else if (time >= 54000.0 & time <= 86400.0)
    {
        temperature = 288;
    }
    begin_f_loop(f, thread)
    {
        F_PROFILE(f, thread, position) = temperature;
    }
    end_f_loop(f, thread)
}
```

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